



Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Impurity and Permutation Importance

WEEK 1 · DAY 4

Created by **Jeff Lenamon**

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Which columns does the forest lean on?

Two standard answers

Impurity importance: `feature_importances_`, free with the fit, sums to 1.

Permutation importance: break one column on unseen rows and measure the drop.

Today's test

Add a column of random numbers. Its worth is known: nothing. Which measure notices?

Before you read a feature importance chart, ask which measure it is and which rows it came from.

Two importances, two different questions

Impurity importance

Each split lowers the Gini impurity of the training rows it splits. Credit the decrease to the split's column, weighted by the rows at the node; add over the tree, average over trees, scale to sum to 1.

From the training rows, during the fit

Permutation importance

Shuffle one column so its values land on the wrong accounts, score the model again, record the drop in validation ROC AUC. Repeat five times and average.

On rows the model never saw

```
table = ensemblekit.importance_table(forest, X_valid, y_valid)
```

One is a byproduct of fitting; the other is an experiment on unseen rows. 18 tests pass today.

A column of random numbers ranks near the top

2nd of 20

noise, by impurity importance

0.0943

its impurity importance

0.0995

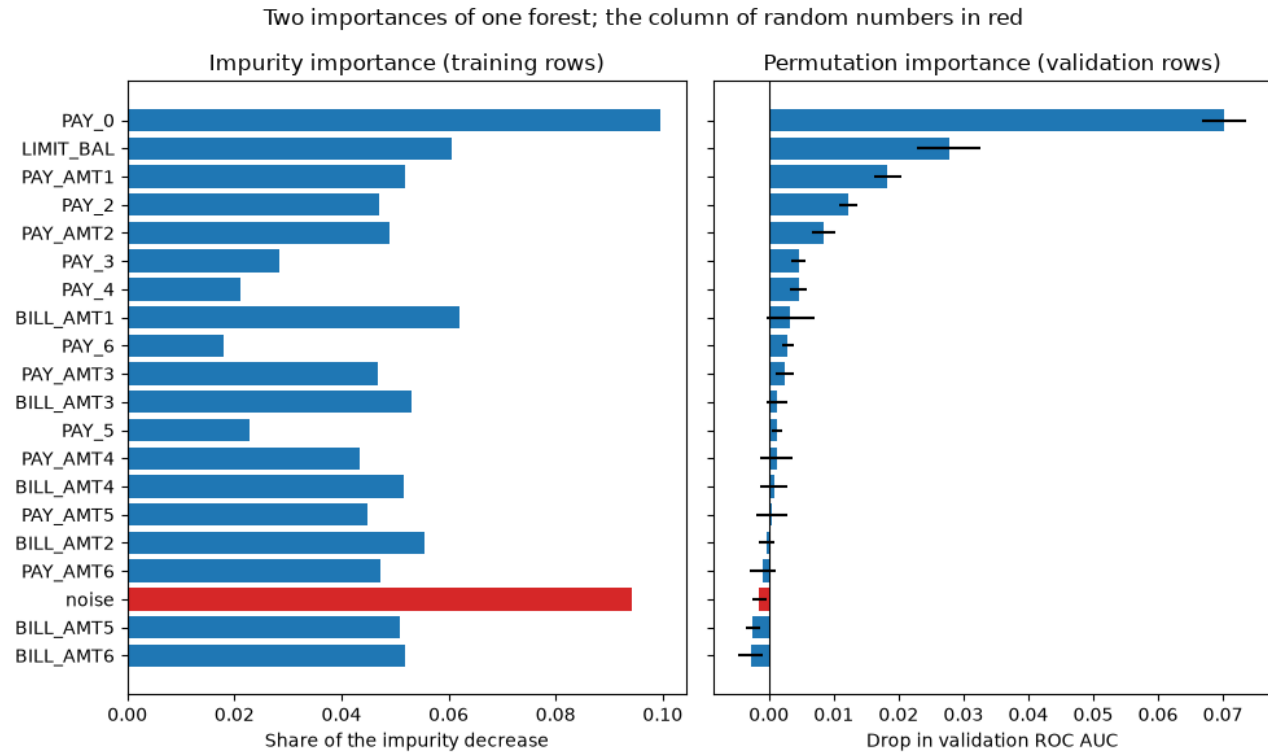
PAY_0, ranked 1st

noise: one normal random number per row, training and validation. Correlation with the outcome -0.0086. The forest of 200 full trees: validation ROC AUC 0.7667, average precision 0.5224.

Impurity rank	Column	Impurity importance
1	PAY_0	0.0995
2	noise	0.0943
3	BILL_AMT1	0.0620
4	LIMIT_BAL	0.0606
5	BILL_AMT2	0.0556

Impurity importance measures how much a column was used to split the training rows.

Permutation importance on validation rows



-0.0015

noise: permutation
importance

18th

its permutation rank, of
20

One shuffle of PAY_0, by hand

Validation ROC AUC 0.7667 as fitted, 0.7051
shuffled: a drop of 0.0616. Same trees, the column
broken.

noise: -0.0015, standard deviation 0.0011 over five shuffles. Zero.

Why impurity importance rewards the noise column

Column	Distinct values
noise	18,000
BILL_AMT1	14,404
PAY_AMT1	5,717
LIMIT_BAL	78
PAY_0	11

Forest	ROC AUC	noise impurity	its rank	noise permutation
Full trees	0.7667	0.0943	2nd	-0.0015
20-row leaves	0.7861	0.0333	10th	0.0005

Deep in a full tree, nodes hold a handful of rows; a column with thousands of values offers thousands of thresholds, and one splits a few defaults off by chance.

Impurity importance is never zero for a column a tree can split on: it cannot say a column is worthless.

Correlated columns share importance

BILL_AMT1 (September 2005) to BILL_AMT6 (April 2005): every pair correlates between 0.80 and 0.95 on the training rows.

-0.0002

six drops, one at a time, added

0.0230

the six shuffled together

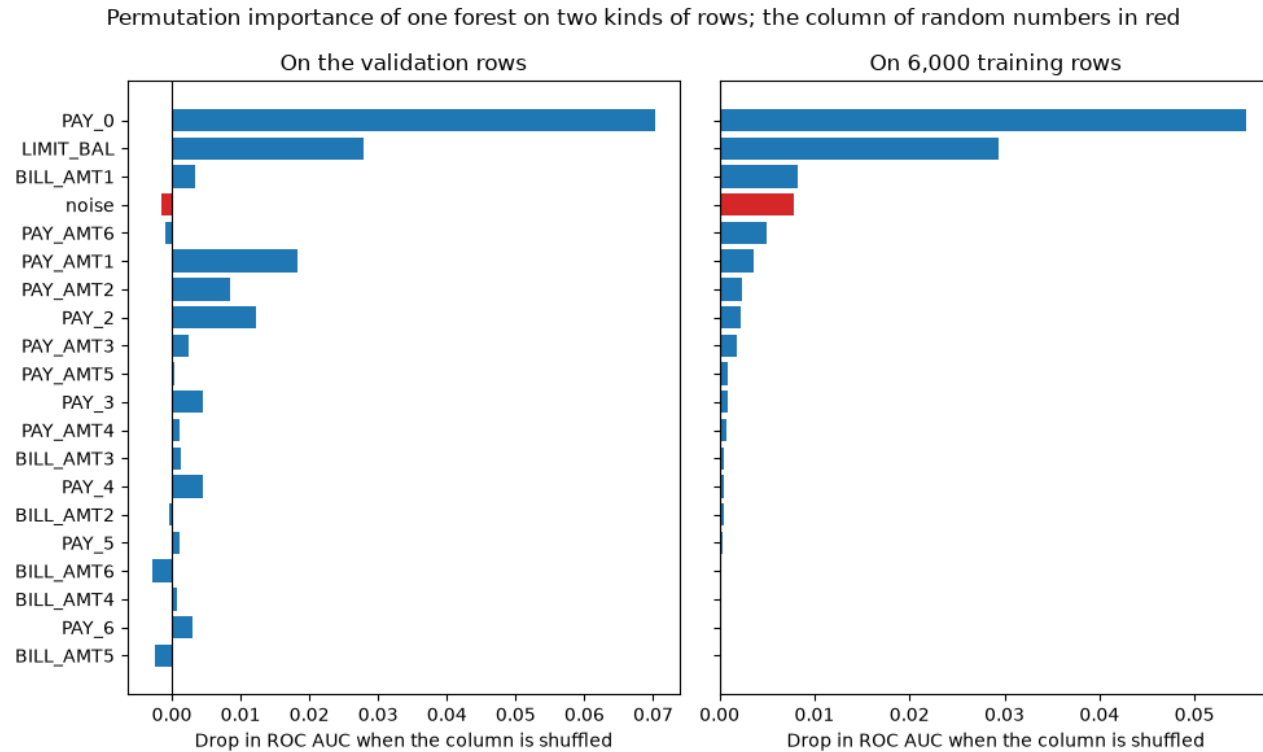
0.3252

their impurity importance, added

```
broken[bills] = X_valid[bills].to_numpy()[order]    # one order of rows for all six
```

Break one and the forest uses its five near-copies: permutation importance of one column is what the model loses when that column alone is broken.

Permutation importance on training rows



0.0079

noise on training rows

-0.0015

noise on validation rows

What it measures there

ROC AUC 1.0000 on those rows. Rank 4th of 20: how much the model used a column to fit them, noise included.

Every time you are shown permutation importance, ask: on which rows?

What to call an importance

Allowed names

impurity importance

permutation importance: the drop in

validation ROC AUC when a column is shuffled

the model leans on; association in this file

Never called

driver, key driver

cause or determinant

risk factor

PAY_0 first on permutation importance: this forest ranks these accounts much worse without it. Nothing about what would happen if an account's payment status changed.

Today: random numbers ranked 2nd by one measure; six related columns near zero by the other.

Excel side by side

Excel side by side (tested in Excel)

`=RANK.EQ(B2,$B$2:$B$21)`

`=RANK.EQ(C2,$C$2:$C$21)`

`=RANK.AVG(B2,$B$2:$B$6001,1)`

`=COUNTIF(A2:A6001,1), =COUNTIF(A2:A6001,0)`

`=(SUMIF(A2:A6001,1,D2:D6001)-H1*(H1+1)/2)/(H1*H2)`

`=CORREL(A2:A18001,B2:B18001)`

noise ranks 2

noise ranks 18

each output's rank, ties averaged

1,327 and 4,673

ROC AUC 0.7667; shuffled 0.7051

0.9511, September with August

No forest, no impurity importance, and no seeded shuffle in Excel: the model's numbers come from Python.

Save the day: HEAD~1

1. Code

mlkit.py, ensemblekit and its tests

2. Log

results.csv: the full-tree forest

3. Log again

add the 20-row-leaf forest

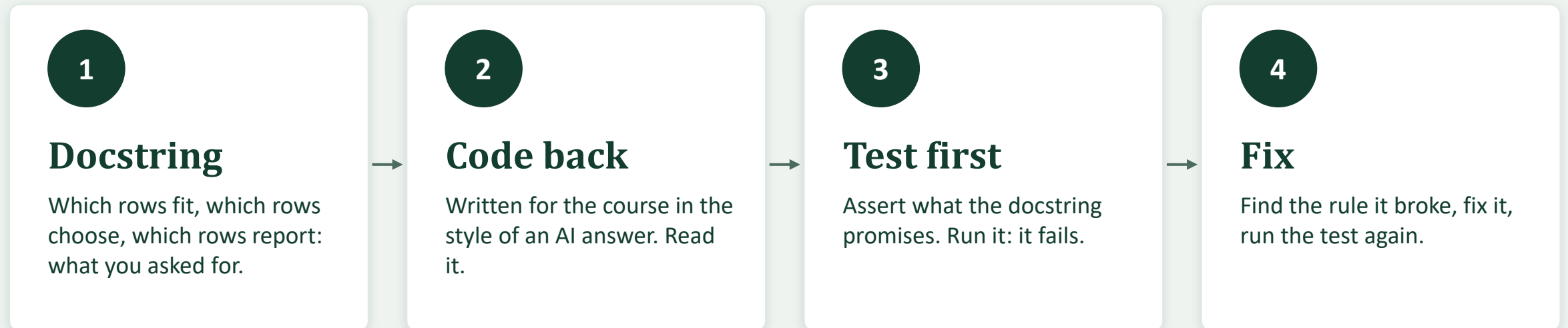
```
!git -C "{work}" show HEAD~1:results.csv
```

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1  
forest_200_full_trees_noise,validation,0.7667,0.5224,0.5000,0.8142,0.6391,0.3670,0.4663
```

HEAD~1:file prints the file one commit back and changes nothing. All-zero baseline 0.7788, never to choose.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: it shuffles the training rows: the noise column looks important on new accounts.

Code that runs is not the same as code that is right.

What you can do now

- Read impurity and permutation importance side by side
- Test a measure with a column of random numbers
- Full trees inflate impurity importance for many-valued columns
- Shuffle correlated columns together
- Ask of permutation importance: on which rows?

Week 1, Day 5 Case study: forests and the spread cross-section